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Studierichting: Informatica

Oefeningenreeks 1B nr. 44.2

$$z^3 - 5(1 + 2i)^2 + 24 - 10i = 0$$

$$\text{Stel } t = z^2 \Rightarrow t^2 - (5 + 10i)t + 24 - 10i = 0$$

De nulpunten zoeken

$$D = (-5 - 10i)^2 - 4 \cdot 1 \cdot (24 - 10i)$$

$$= 25 + 100i - 100 - 96 + 40i$$

$$= -171 + 140i$$

Discriminant is een complex getal.

$$-171 + 140i = x^2 + 2xyi - y^2 \Rightarrow \begin{cases} x^2 - y^2 = -171 \\ 2xy = 140 \Rightarrow x^2(-y^2) = 4900 \end{cases}$$

$$\lambda^2 + 171\lambda - 4900 = 0$$

Berekening van nulpunten:

$$D = 171^2 - 4 \cdot 1 \cdot (-4900) = 48841 \Rightarrow \sqrt{100} = 221$$

$$\lambda_{1,2} = \frac{-171 \pm 221}{2}$$

$$\lambda_1 = 25 \text{ en } \lambda_2 = -196$$

$$x = \pm 5 \text{ en } y = \pm 14i$$

xy is positief $\Rightarrow$ vierkantswortel van eerste discriminant: $5 + 14i$ of $-5 - 14i$

$$t_{1,2} = \frac{5 + 10 \pm (5 + 14i)}{2}$$

$$t_1 = 5 + 12i \text{ en } t_2 = -2i$$

$$t = z^2 \Leftrightarrow z = \sqrt{t} :$$

Voor $t = 5 + 12i$

$$5 + 12i = x^2 + 2xyi - y^2 \Rightarrow \begin{cases} x^2 - y^2 = 5 \\ 2xy = 12 \Rightarrow x^2(-y^2) = 36 \end{cases}$$

$$\lambda^2 - 5\lambda - 36 = 0$$

Berekening van nulpunten:

$$D = 25 - 4 \cdot 1 \cdot (-36) = 169 \Rightarrow \sqrt{169} = 13$$

$$\lambda_{1,2} = \frac{5 \pm 13}{2}$$

$$\lambda_1 = 9 \text{ en } \lambda_2 = -4$$

$$x = \pm 3 \text{ en } y = \pm 2$$

$$xy \text{ is positief} \Rightarrow z_1 = 3 + 2i \text{ en } z_2 = -3 - 2i$$

Voor $t = -2i$

$$-2i = x^2 + 2xyi - y^2 \Rightarrow \begin{cases} x^2 - y^2 = 0 \\ 2xy = -2 \Rightarrow x^2(-y^2) = 1 \end{cases}$$

$$\lambda^2 - 1 = 0$$

$$\lambda^2 = 1$$

$$\lambda_{1,2} = \pm 1$$

$$x = \pm 1 \text{ en } y = \pm 1$$

$$xy \text{ is negatief} \Rightarrow z_3 = 1 - i \text{ en } z_4 = -1 + i$$

References